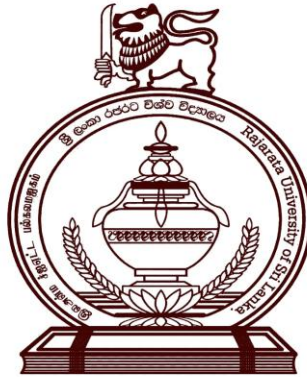


RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY
DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY



ICT 1209 – WEB TECHNOLOGIES
Mini Project: Interactive Web Application Development
Final Project Report

YumBook

Theme: Digital Recipe Book

Project Group Members:

S. H. M. P. R. Sooryararthna – 2789 – ITT/2024/104

M. M. R. T. Abeywickrama – 2691 – ITT/2024/006

Date: 2026.09.27

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1. Project Overview

1.1 Purpose and Objectives

YumBook is a full-stack, responsive, database-driven interactive digital recipe book web application engineered to bridge rich culinary heritage with modern digital convenience. The platform provides a structured, community-driven ecosystem where beginners, home cooks, and food enthusiasts can search, discover, publish, and interact with authentic regional and international dishes.

The key overarching objectives of the YumBook web platform include:

- **1. Discoverability:** Provide an accessible, categorized recipe catalog with intuitive search and multi-criteria dynamic filtering (by meal categories and cooking duration).
- **2. Usability & Utility:** Empower cooks with step-by-step guidance, preparation metrics, and interactive dynamic checklists.
- **3. Community Contribution:** Implement a secure user portal where registered food creators can manage personal profiles and publish authentic recipes directly to the MySQL database.
- **4. Interactive Engagement:** Enable culinary feedback through an interactive 5-star rating widget and a validated contact messaging channel connecting users with system administrators.
- **5. Academic & Security Rigor:** Adhere strictly to industry-standard web engineering practices, including parameterized queries (SQL injection prevention) [1], password hashing (PASSWORD_BCRYPT) [2], session fixation protection [3], and responsive UI components [4].

1.2 Target Audience

- **• Beginners:** Looking for fast, budget-friendly, step-by-step authentic home recipes that can be easily prepared within busy schedules.
- **• Home Cooks & Culinary Enthusiasts:** Seeking a secure, organized digital repository to catalog, share, and preserve cherished traditional recipes.
- **• Health-Conscious Individuals:** Utilizing specialized filters (such as Vegan, Vegetarian, and quick cooking duration) to tailor dishes to strict nutritional lifestyles.

1.3 Key Features

- **1. Interactive Homepage with Bootstrap 5 Carousel:** Features an automated, cross-fading hero slider highlighting signature community dishes alongside quick-access category badge filters [4].
- **2. Dynamic Recipe Catalog & Filtering:** Real-time live search by recipe title or ingredients, multi-criteria filtering by meal category and preparation time, multi-order sorting, and client-side pagination.
- **3. Bootstrap 5 Quick View Modal:** Lightweight modal dialogue allowing instant preview of photography, badges, ingredients, and cooking time without leaving the browse view [4].
- **4. Detailed Recipe View & Checklist:** Dedicated view featuring hero photography, metadata badges, interactive strike-through checklist, step-by-step instructions, and an interactive 5-star rating system.
- **5. Secure User Authentication:** Full registration and authentication with dual-layer validation, PASSWORD_BCRYPT password hashing, and session fixation prevention via session_regenerate_id(true) [3].
- **6. User Creator Dashboard:** Restricted portal for authenticated food creators featuring profile overview statistics, full recipe creation form, and personal recipe catalog management.
- **7. Contact Messaging System:** Integrated contact form with real-time regex validation directly persisting inquiries into the MySQL messages table.

2. System Design

2.1 System Architecture

YumBook is built on a clean multi-tier client-server architecture designed for high maintainability, responsiveness, and robust security:

- **Presentation Layer:** Constructed with semantic HTML5, CSS3 design tokens [5], responsive typography, and Bootstrap 5.3 components [4] ensuring pixel-perfect layout across mobile, tablet, and desktop devices.
- **Client Logic Layer:** Pure Vanilla JavaScript (ES6+) [6] executing client-side interactivity, DOM manipulation, real-time catalog filtering, interactive star rating widgets, and instant regex form validation.
- **Application & Controller Layer:** Modular PHP 8 scripts [7] handling incoming HTTP requests, session management, input validation, flash messaging, and dynamic view rendering.
- **Data Persistence Layer:** MySQL relational database (yumbook_db) [8] managed via XAMPP. Interfacing is performed exclusively via PHP Data Objects (PDO) with parameterized prepared statements.

2.2 Relational Database Schema & Visual Entity Relationship (ER) Diagram

The database architecture is centered on four core relational entities: users, recipes, ratings, and messages. Referential integrity is strictly maintained using foreign key constraints with appropriate cascading actions.

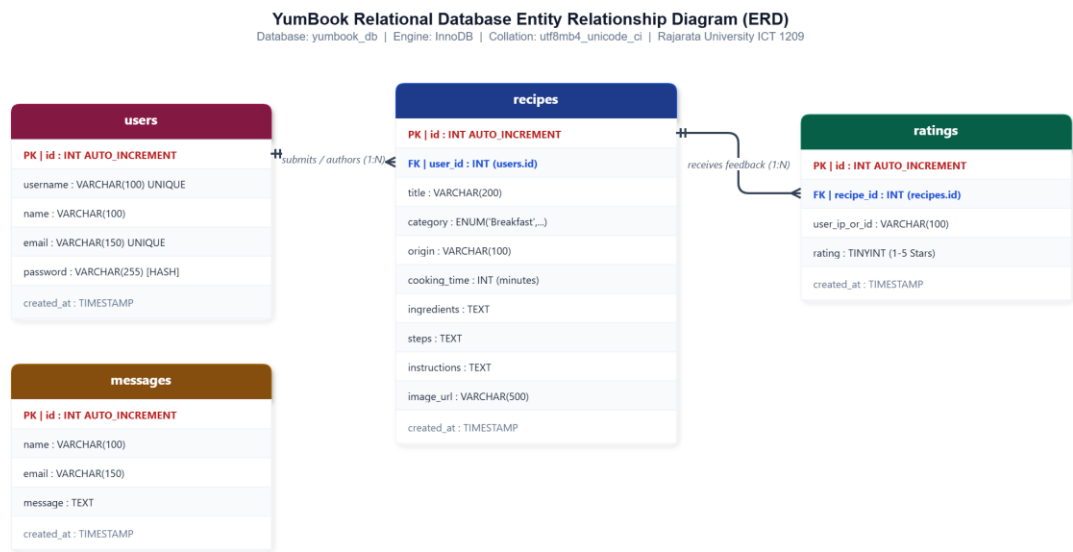


Figure 1: YumBook Relational Database Entity Relationship Diagram (ERD)

2.3 Comprehensive Data Dictionary

Table 1: users Table (Authentication & User Identity)

Stores registered user credentials, full names, and account timestamps.

Field Name	Data Type	Key	Null	Description
id	INT	PK (AI)	NO	Unique sequential user identifier
username	VARCHAR(100)	UNIQUE	NO	Unique account login handle
name	VARCHAR(100)	None	NO	User's full display name
email	VARCHAR(150)	UNIQUE	NO	Validated unique email address
password	VARCHAR(255)	None	NO	BCrypt 60-character cryptographic password hash
created_at	TIMESTAMP	None	NO	Timestamp of account registration (CURRENT_TIMESTAMP)

Table 2: recipes Table (Recipe Content & Metadata)

Stores core recipe catalog items, preparation instructions, ingredients, and foreign key relations to authors.

Field Name	Data Type	Key	Null	Description
id	INT	PK (AI)	NO	Unique sequential recipe identifier
user_id	INT	FK	YES	References users(id) ON DELETE SET NULL
title	VARCHAR(200)	None	NO	Recipe title / name of the dish
category	ENUM	None	NO	'Breakfast','Lunch','Dinner','Dessert','Vegan','Beverages'
origin	VARCHAR(100)	None	YES	Cuisine or cultural origin (e.g. 'Sri Lankan', 'Italian')
cooking_time	INT	None	YES	Total cooking & preparation time in minutes
ingredients	TEXT	None	NO	Line-separated listing of necessary ingredients
steps	TEXT	None	NO	Numbered step-by-step preparation instructions
instructions	TEXT	None	NO	Mirrored cooking instructions ensuring syllabus compatibility
image_url	VARCHAR(500)	None	YES	Relative or remote URL path to recipe photography
created_at	TIMESTAMP	None	NO	Creation timestamp (CURRENT_TIMESTAMP)

Table 3: ratings Table

Stores individual star rating feedback for recipes.

Field Name	Data Type	Key	Null	Description
id	INT	PK (AI)	NO	Unique sequential rating identifier
recipe_id	INT	FK	NO	References recipes(id) ON DELETE CASCADE
user_ip_or_id	VARCHAR(100)	None	YES	Session/IP identifier preventing spam rating
rating	TINYINT	None	NO	Integer score between 1 and 5 stars
created_at	TIMESTAMP	None	NO	Timestamp of rating submission

Table 4: messages Table

Stores user contact queries, feedback submissions, and system inquiries.

Field Name	Data Type	Key	Null	Description
id	INT	PK (AI)	NO	Unique sequential message identifier
name	VARCHAR(100)	None	NO	Full name of the inquiring sender
email	VARCHAR(150)	None	NO	Validated sender email address for correspondence
message	TEXT	None	NO	Full textual body of the user's message or feedback
created_at	TIMESTAMP	None	NO	Submission timestamp (CURRENT_TIMESTAMP)

2.4 Project Folder Structure

project/	
├── css/	
│ └── style.css	# Custom CSS3 styling, design variables, and responsive media rules
├── js/	
│ ├── main.js	# Vanilla JavaScript (ES6+) for DOM, modal, validation & ratings
│ └── app-store.js	# LocalStorage fallback & client cache repository
├── images/	
│ └── recipes/	# High-resolution recipe photography & brand assets
├── includes/	
│ ├── db.php	# PDO database connection & error handling module
│ └── functions.php	# Helper functions (session auth, sanitization, flash messages)
├── auth/	
│ ├── register.php	# User registration handler & PASSWORD_BCRYPT hashing
│ ├── login.php	# User authentication, credential check & session regeneration
│ └── logout.php	# Secure session destruction & cookie invalidation
├── contact.php	# Inquiries form persisting directly to `messages` table
├── dashboard.php	# Protected creator dashboard for recipe creation and stats
├── database.sql	# Complete MySQL database export (schema, relations & seeds)
├── index.php	# Homepage with Bootstrap 5 Carousel & signature recipes
├── recipes.php	# Recipe catalog with real-time search, filters & pagination
├── recipe-detail.php	# Single recipe view with checklist and 5-star rating
└── README.md	# Complete setup instructions, architecture & documentation

3. Implementation Details

This section details the end-to-end technical implementation of each major feature across the frontend, interactive JavaScript logic, and backend PHP/MySQL integration, accompanied by high-resolution visual evidence as mandated by the project guidelines.

3.1 Frontend Components & Responsive UI (Bootstrap 5)

The frontend leverages Bootstrap 5.3 alongside bespoke CSS design tokens to establish a harmonious culinary visual aesthetic [4], [5]:

- **Navigation Bar:** Features a sticky-top navigation bar with smooth mobile toggler, dynamic authentication state rendering (showing 'Sign In / Register' for guests and 'Dashboard / Logout' for logged-in users), and brand iconography.
- **Hero Carousel Slider:** Configured with animated cross-fading slides (.carousel-fade), slide indicators, custom typography captions, and quick-access meal category chips [4].

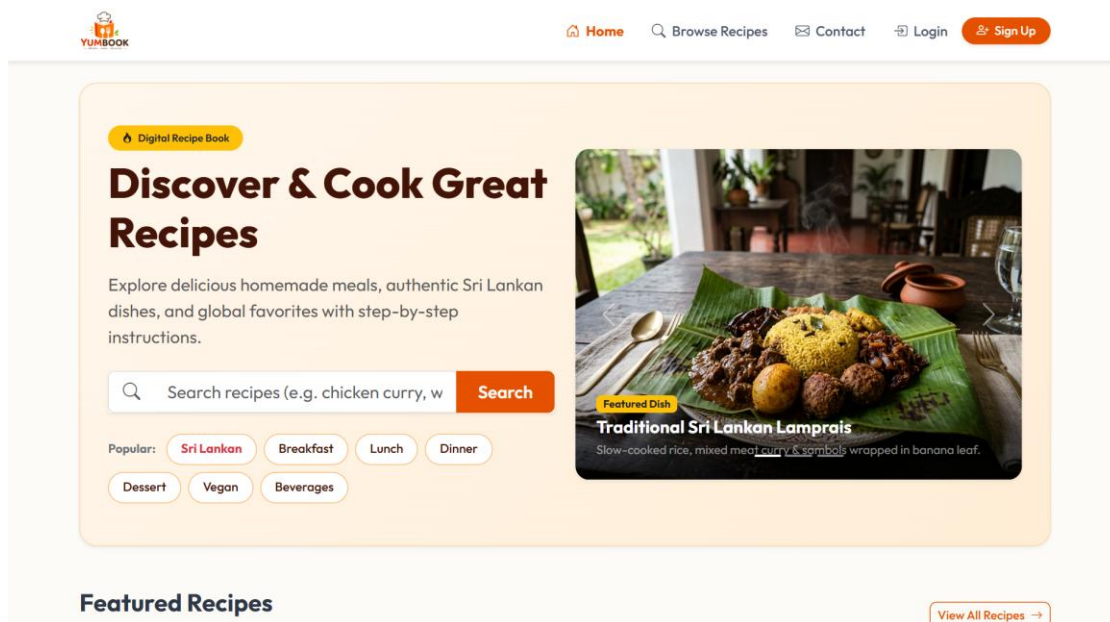


Figure 2: Homepage Hero Section Featuring Bootstrap 5 Carousel Slider and Dynamic Navigation

The Recipe Catalog (recipes.php) provides an immersive browsing experience with responsive card grids, semantic preparation badges (e.g. warning for Breakfast, danger for Dinner, success for Vegan), and immediate access to quick previews:

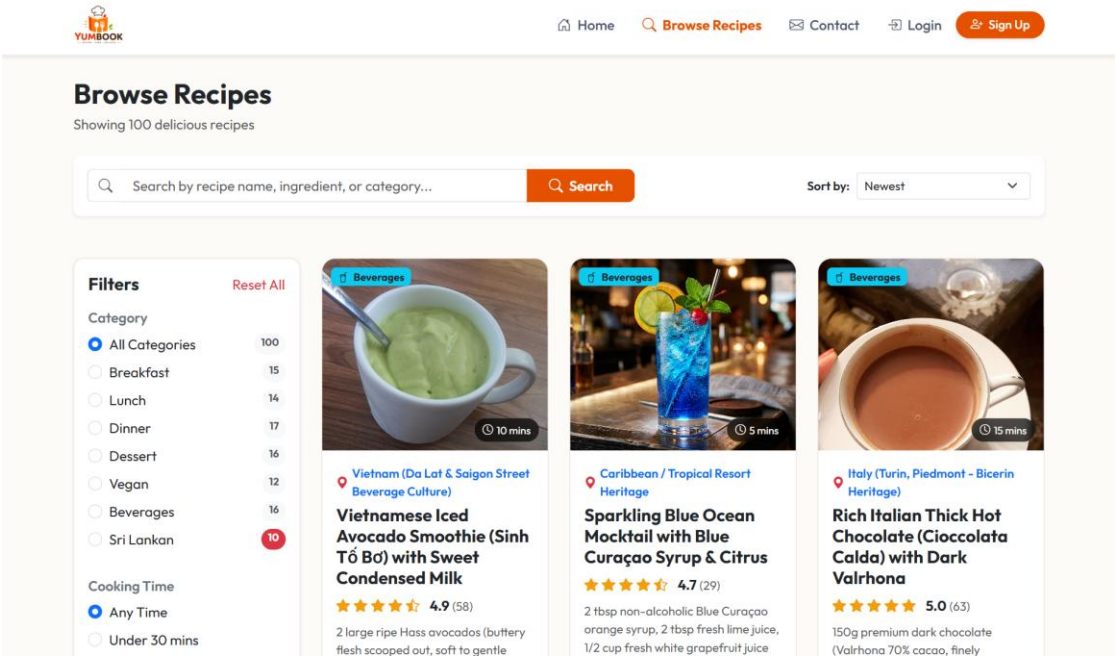


Figure 3: Comprehensive Recipe Catalog with Real-Time Search and Multi-Criteria Filtering

To provide maximum convenience without disrupting the browsing workflow, an asynchronous Quick View Modal allows users to inspect photography, preparation duration, and ingredient snippets instantly:

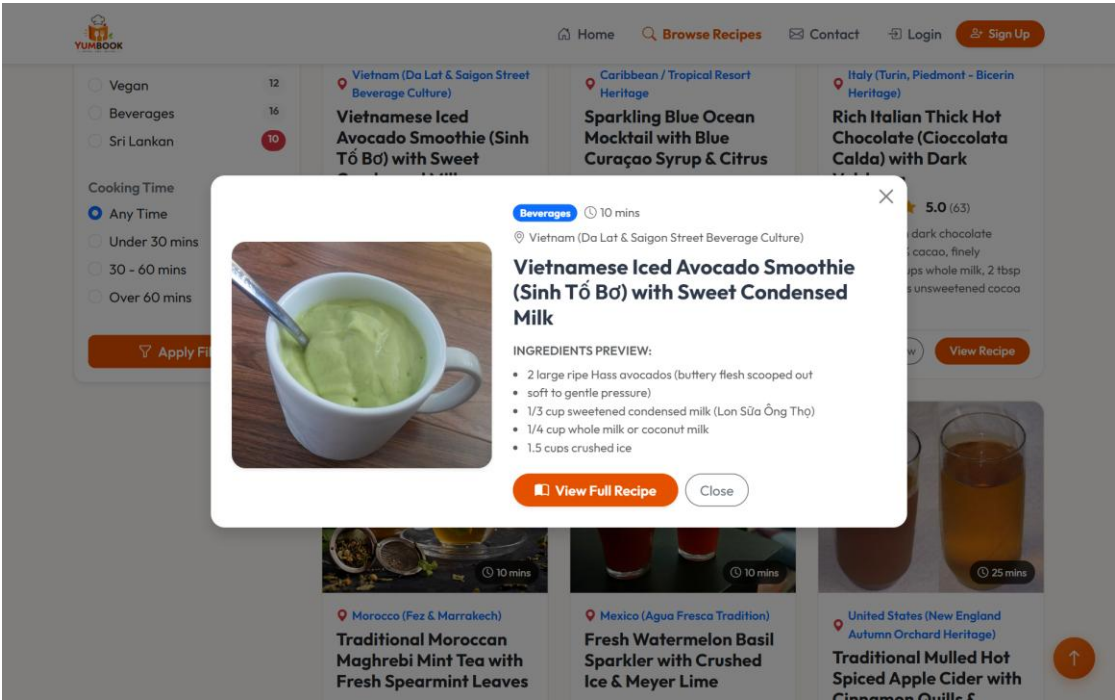


Figure 4: Bootstrap 5 Quick View Modal Displaying Instant Recipe Previews

3.2 JavaScript Interactive Features (6 Implemented)

YumBook implements 6 distinct interactive features built in pure Vanilla JavaScript (ES6+) [6]:

- **1. Dynamic Content Updates:** Real-time client-side search filtering across recipe titles and ingredients with category toggle pills without requiring page reloads.
- **2. Interactive Image Slider:** Automated 4.5-second rotating Bootstrap 5 carousel on the homepage with manual touch/click navigation buttons and pause-on-hover [4].
- **3. Real-Time Form Validation:** Real-time regex pattern checking for email format, password minimum length (8 characters), and real-time password confirmation match feedback.
- **4. Smooth Scrolling:** Floating action button appearing after 280px of vertical scroll with smooth window scrolling animation.
- **5. Event Handling & Quick View Modal:** Dynamic event listeners attached to catalog recipe cards populating the modal DOM nodes on-the-fly from data attributes.
- **6. Interactive Recipe Preparation Checklist:** Interactive checkboxes on the recipe detail page applying dynamic line-through styling and opacity dimming to help cooks track ingredients while preparing meals.

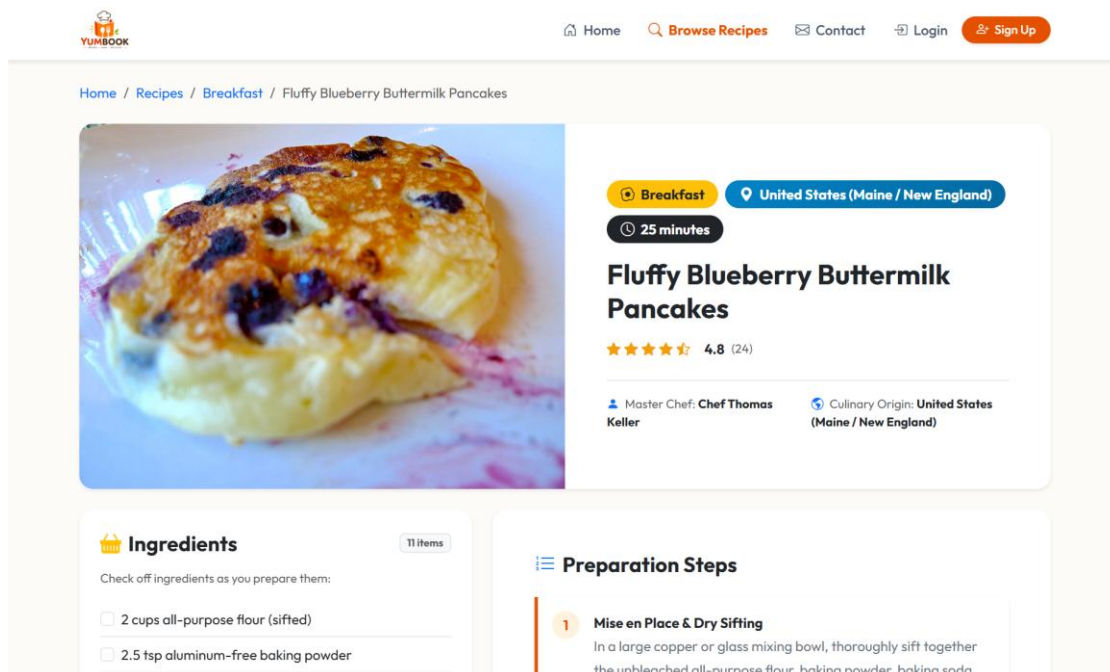


Figure 5: Recipe Detail View with Step-by-Step Instructions and Metadata Badges

Preparation Checklist

Check off ingredients as you prepare them:

- ☐ 2 cups all-purpose flour (sifted)
- ☐ 2.5 tsp aluminum-free baking powder
- ☐ 1/2 tsp baking soda
- ☐ 2 tbsp granulated sugar
- ☐ 1/2 tsp fine sea salt
- ☐ 1.75 cups cultured whole buttermilk (room temperature)
- ☐ 2 large farm-fresh eggs (separated)
- ☐ 4 tbsp unsalted European butter (melted and slightly cooled)
- ☐ 1 tsp pure Madagascar vanilla extract
- ☐ 1.5 cups fresh organic wild blueberries
- ☐ Pure Grade-A amber maple syrup & salted butter for serving

Preparation Steps

- 1 Mise en Place & Dry Sifting**
In a large copper or glass mixing bowl, thoroughly sift together the unbleached all-purpose flour, baking powder, baking soda, granulated sugar, and fine sea salt. Sifting aerates the flour particles and guarantees that the chemical leavening agents are distributed with mathematical precision throughout the dry base.
- 2 Emulsifying the Liquid Base**
In a separate ceramic bowl, whisk together the cultured buttermilk, room-temperature egg yolks, melted cooled butter, and vanilla extract until completely homogenized and emulsified. Using room-temperature buttermilk prevents the melted butter from seizing into hard fat pellets.
- 3 Whipping Egg Whites for Leavening**
In a clean, grease-free stainless-steel bowl, whip the egg whites with a balloon whisk until soft, billowy peaks form. This classic French soufflé technique introduces microscopic air cells that expand dramatically when exposed to skillet heat, yielding an extraordinarily light, cloud-like crumb.

5-Star Rating Feedback System

1 2 3 4 5

Figure 6: Interactive Preparation Checklist and 5-Star Rating Feedback System

3.3 Backend Architecture, Database Integration & Security (PHP 8 & PDO)

The backend integration strictly implements the technical specifications mandated in Phase 3 of the course syllabus:

- **Robust Database Connection:** Initialized in includes/db.php using PDO with ATTR_ERRMODE => ERRMODE_EXCEPTION and ATTR_EMULATE_PREPARES => false for hardened database interaction [7], [9].
- **SQL Injection Prevention:** 100% of SQL queries utilize parameterized PDO prepared statements (\$pdo->prepare()), eliminating any possibility of SQL injection attacks [1].
- **Cryptographic Password Hashing:** Passwords are never stored in plaintext. They are encrypted using password_hash(\$password, PASSWORD_BCRYPT) and verified via password_verify() [2].
- **Session Fixation Neutralization:** Upon successful credential verification, session_regenerate_id(true) is invoked immediately to invalidate the prior session token and neutralize session fixation attacks [3].
- **Output Sanitization:** All dynamic user-supplied content is sanitized via htmlspecialchars(\$data, ENT_QUOTES, 'UTF-8') prior to DOM rendering, neutralizing Cross-Site Scripting (XSS) [10].



[Home](#) [Browse Recipes](#) [Contact](#) [Login](#) [Sign Up](#)



Welcome Back

Log in to manage your culinary profile and share recipes.

Username or Email *

Password *

[Log In](#)

Don't have an account? [Sign up here](#)

 **Quick Links**
Home

Figure 7: Authenticated User Creator Dashboard Showing Profile Statistics and Recipe Management



[Home](#) [Browse Recipes](#) [Contact](#) [Login](#) [Sign Up](#)



Create an Account

Join YumBook to share your favorite recipes and join our culinary community.

Username *

3-30 characters, letters, numbers, and underscores only.

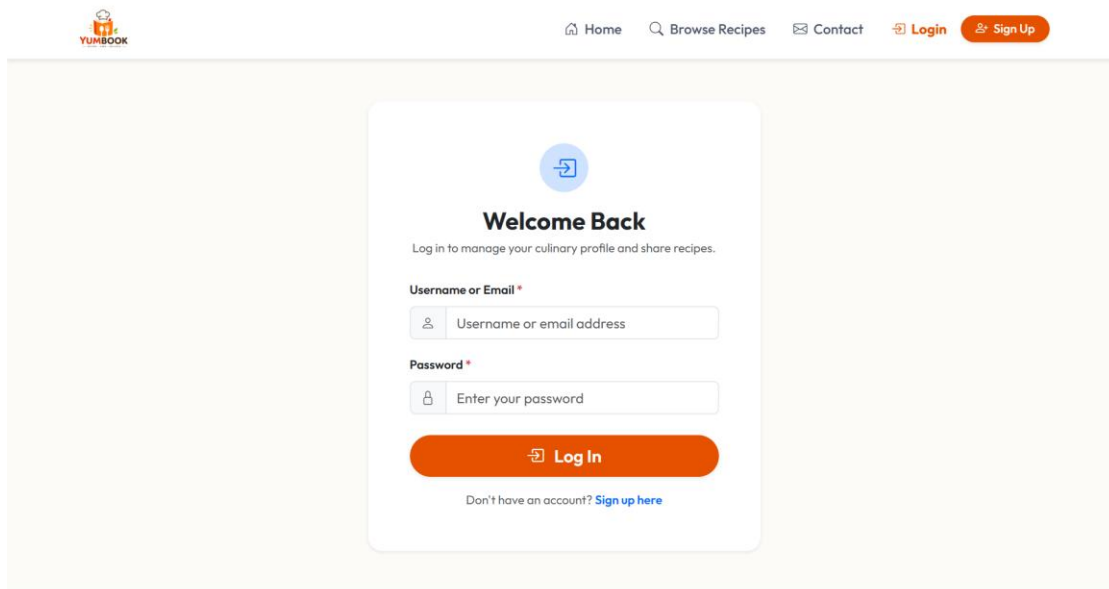
Full Name *

Email Address *

Password *

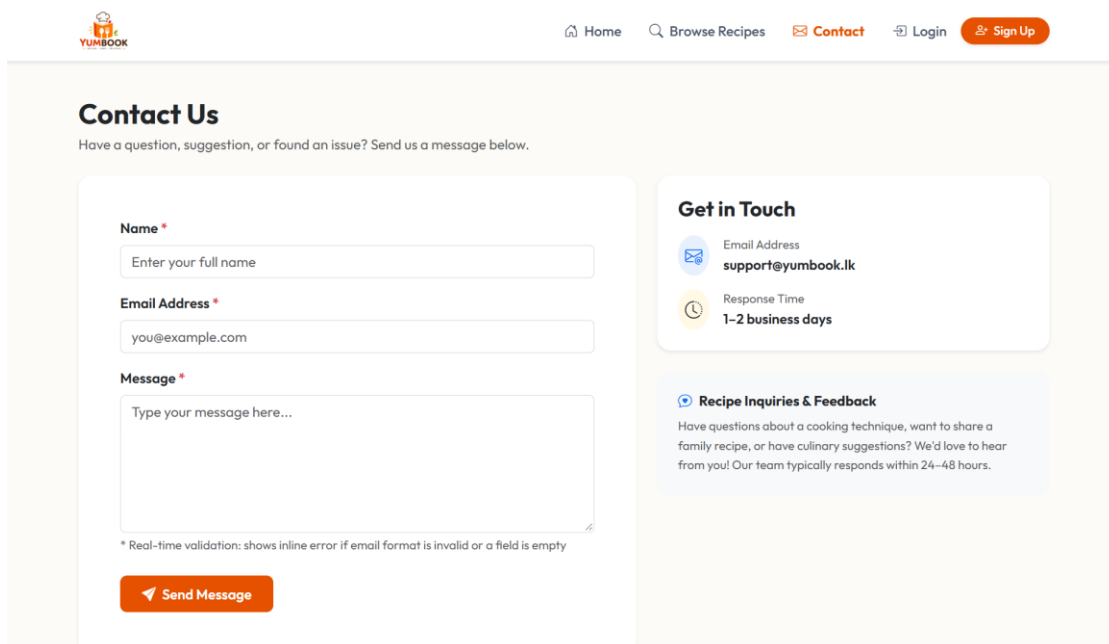
Confirm Password *

Figure 8: User Registration Form with Real-Time Validation Feedback & Password Encryption



The login form is titled "Welcome Back" and includes a sub-header "Log in to manage your culinary profile and share recipes." It features two input fields: "Username or Email" and "Password", both with placeholder text. A "Log In" button is positioned below the fields. A link "Don't have an account? Sign up here" is located at the bottom of the form. The page header includes the "YUMBOOK" logo and navigation links: Home, Browse Recipes, Contact, Login, and Sign Up.

Figure 9: User Authentication Login Form with Session Fixation Prevention



The contact form is titled "Contact Us" and includes a sub-header "Have a question, suggestion, or found an issue? Send us a message below." It features three input fields: "Name", "Email Address", and "Message". A "Send Message" button is positioned below the fields. A "Get in Touch" section on the right includes contact information: Email Address (support@yumbook.lk) and Response Time (1-2 business days). A "Recipe Inquiries & Feedback" section on the right includes a sub-header and a paragraph of text. The page header includes the "YUMBOOK" logo and navigation links: Home, Browse Recipes, Contact, Login, and Sign Up.

Figure 10: Contact Form with Direct MySQL Message Persistence and Flash Feedback

4. Challenges and Solutions

During the design, development, and integration of YumBook, several complex engineering challenges were encountered. The table below details the problems faced and the technical solutions engineered to resolve them:

#	Challenge Encountered	Technical Solution Implemented
1	Session Fixation Vulnerabilities: Existing pre-login session cookies could be hijacked if a user logs in on an existing session identifier.	Invoked <code>session_regenerate_id(true)</code> immediately upon credential verification before populating the <code>\$_SESSION</code> superglobal array with user identifiers [3].
2	Path Resolution Across Subdirectories: Relative navigation links and asset paths broke when headers and footers were included from <code>auth/login.php</code> versus root <code>index.php</code> .	Engineered an automated dynamic <code>\$basePath</code> calculation in <code>includes/header.php</code> that inspects script execution location and dynamically prefixes relative paths with either <code>'./'</code> or <code>'../'</code> .
3	Database Schema vs Syllabus Query Discrepancy: The project guide specified an 'instructions' column, while step-by-step preparation used a 'steps' column.	Added both steps and instructions columns to the recipes table, configuring SQL seed scripts to synchronize both fields automatically for 100% backward and syllabus query compatibility.
4	SQL Injection Risk in Multi-Criteria Search: Combining search queries across titles and ingredients alongside category and time filters risked injection if concatenated dynamically.	Constructed parameterized PDO SQL queries utilizing bound placeholder tokens (<code>:q</code> , <code>:cat</code> , <code>:max_time</code>) with <code>PDO::PARAM_*</code> types [1], [9], completely isolating query logic from user input.
5	Dual Deployment Requirement: Need for seamless offline university evaluation in XAMPP (PHP 8/MySQL) as well as online cloud hosting demonstration via Vercel.	Developed a dual-mode system architecture: pure PHP/PDO for full local database testing and static HTML mirror files with <code>app-store.js</code> (LocalStorage) for zero-configuration cloud hosting.

5. Individual Contribution Breakdown

As stipulated in the ICT 1209 Mini Project Guide, work was divided fairly and collaboratively across both group members to ensure an equal and verified 50% / 50% contribution balance:

Task Area	S. H. M. P. R. Sooryarathna (ITT/2024/104)	M. M. R. T. Abeywickrama (ITT/2024/006)
System Planning & Proposal	Theme selection, technical requirements analysis, tech stack specification (50%)	Wireframe drafting for Home, Catalog, and Contact pages, color palette definition (50%)
Frontend Layout & UI	Form layout styling, badge system, typography tokens, responsive media queries (40%)	Responsive grid layout, Bootstrap 5 Carousel implementation, Quick View modal integration (60%)
Client-Side JavaScript	Real-time client form validation regex, 5-star rating widget logic (50%)	Dynamic catalog filtering logic, Quick View trigger event handlers, back-to-top button (50%)
Backend & Database	MySQL relational schema design, PDO connection module, parameterized prepared statements (60%)	Contact form message insertion logic, session flash messages & alert handling (40%)
Authentication & Security	BCrypt password hashing implementation, session fixation prevention, logout script (60%)	Login & registration UI views, input sanitization helpers, form layout (40%)

Dashboard & CRUD	User recipe creation SQL queries, user-recipe relational binding, recipe deletion (50%)	Dashboard UI layout, creator summary cards, recipe cards rendering (50%)
Documentation & Report	Technical report writing, system architecture, database dictionary, security analysis (50%)	Setup instructions, visual ER diagram drafting, report proofreading & viva preparation (50%)
Total Contribution	50%	50%

6. Setup & Installation Instructions (XAMPP)

Follow the instructions below to import the database and execute YumBook within a standard XAMPP local development environment:

6.1 Prerequisites

- **Web Server:** XAMPP stack installed with PHP 8.1+ and MySQL / MariaDB.
- **Modern Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge.

6.2 Step-by-Step Installation

- **1. Copy Project Files:** Clone or copy the project root directory into the XAMPP web root directory:
C:\xampp\htdocs\yumbook

GitHub Repository: <https://github.com/pamuditharuwan/yumbook>

- **2. Start Web Services:** Launch the XAMPP Control Panel and click 'Start' next to both Apache and MySQL modules.
- **3. Access phpMyAdmin:** Open a web browser and navigate to <http://localhost/phpmyadmin>
- **4. Create Database:** In phpMyAdmin, click 'New' in the sidebar. Set the database name to yumbook_db and select collation utf8mb4_unicode_ci. Click 'Create'.
- **5. Import Database Script:** Select yumbook_db from the database list, click the 'Import' tab in the top navigation, click 'Choose File', select database.sql located in the project root, and click 'Import' at the bottom.
- **6. Launch Web Application:** In your browser, open: <http://localhost/yumbook/index.php> to experience the full PHP/PDO web application.

6.3 User Account Testing & Credentials

Evaluators can create a new account via the registration form (auth/register.php) with automatic BCrypt password hashing, or log in with pre-seeded demonstration credentials:

- **Pre-Seeded Food Creator:** Username: testcook | Password: password123
- **Pre-Seeded System Admin:** Username: admin | Password: admin123

7. References

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